

Synchronizing Dzyga's Automata by a Cyclic Retract

An explicit all-parameter reset construction,
a nearby obstruction, and certified pair distances

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Abstract

We study the binary automata $\mathcal{A}_k$ on $n = 3k + 6$ states that occur in Conjecture 8 of Szykuła's 2026 survey of synchronizing automata. We give a uniform proof that every $\mathcal{A}_k$, $k \geq 1$, is synchronizing. The construction uses the word ab : its unique recurrent component is a cycle of length $r = k + 4$, and $p = (ab)^r$ is an idempotent retraction onto this cycle. Two projected letter actions become a reflection and a reflection with one defect. Their composition produces an elementary merger, from which we construct a reset word of length

$$8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16.$$

This resolves the synchronization clause for the figure-defined family, not the conjectured exact compress-with-another threshold. We also prove that shortening the tail by one state produces a nonsynchronizing family with a dihedral permutation quotient. The known upper bound $4k + 8$ for compressing q_0 is reproved and attributed to prior work. Independent integer certificates establish the exact threshold and all pair distances for $1 \leq k \leq 200$. The universal matching lower bound remains unproved here. Complete proofs, source-model conventions, executable checks, and verification limitations are included.

Scope of the result

Proved for every parameter: a reset word for $\mathcal{A}_k$; a quadratic reset-length upper bound; a permutation-quotient obstruction for the one-state-shorter comparison family.

Not proved for every parameter: that $4k + 8$ is the *minimum* length merging q_0 with another state, or the proposed exact formula for the pairwise merging diameter. The supplied finite certificates do not replace these missing uniform lower-bound proofs.

Research status. This is a new research draft with an explicit, independently checkable argument, not a peer-reviewed publication or a proof-assistant formalization. Novelty is assessed against the sources identified below, not asserted against all possible prior literature.

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1 The problem selected and the result obtained

1.1 A concrete open question in automata theory

A deterministic finite automaton transforms its state each time it reads a letter. A *reset word* sends every possible initial state to the same final state. The problem of finding such a word is therefore a question about a language of words, not about choosing an initial or an accepting state. Our automata are more precisely complete deterministic *semiautomata*; the shorter conventional name will be used throughout.

Szykuła's survey [1, Section 2.3, Figure 3 and Conjecture 8] records a family $\mathcal{A}_k$ on $3k + 6$ states, credited there to Dżyga's 2018 master's thesis. Its conjecture combines universal synchronizability with an exact shortest-word assertion for one distinguished state. We select the first clause as the main research problem:

For the family $\mathcal{A}_k$ defined in [section 3](#), does a reset word exist for every integer $k \geq 1$?

The advantage of this target is that a positive answer can be made entirely explicit. It is not necessary to characterize every shortest word, and it is not necessary to infer a universal pattern from a finite search. One can instead identify transformations induced by a few parameterized words and then compose them to obtain a constant transformation.

An intervening online research manuscript [2] already proves the all-parameter upper bound $4k + 8$ for merging the distinguished state and corrects the displayed witness word. It reports exact finite checks through $k = 60$, but explicitly leaves universal synchronizability open. Those upper bounds and the correction are *prior results*, not contributions claimed by the present draft. We prove synchronization by a different structural argument and extend the independently reproduced finite checks to $k = 200$.

The original thesis has not been examined for this article. Thus historical attribution to it is through the survey, and no assertion is made about the precise wording or contents of that thesis. The available source record supports treating the selected clause as open, but it cannot guarantee absolute priority for the argument given here.

1.2 Main theorem

Words act on the right and are read from left to right: $q(uv) = (qu)v$. A product of words below is a literal concatenation, even when we use the same notation for the state transformations that the words induce.

Theorem 1.1 (Explicit synchronization). *Let $k \geq 1$ and let $\mathcal{A}_k$ be the complete binary automaton of [definition 3.1](#). Put*

$$r = k + 4, \quad c = ab, \quad p = c^r, \quad e = apbpc^{r-1}. \quad (1.1)$$

Then the word

$$R_k = p(ec^{r-1})^{r-2}e \quad (1.2)$$

sends every state to q_2 . In particular, $\mathcal{A}_k$ is synchronizing and

$$\text{rt}(\mathcal{A}_k) \leq |R_k| = 8r^2 - 10r + 4 = 8k^2 + 54k + 92 = \frac{8}{9}n^2 + \frac{22}{3}n + 16, \quad n = 3k + 6. \quad (1.3)$$

The notation $\text{rt}(\mathcal{A}_k)$ denotes the length of a *shortest* reset word. The theorem supplies an upper bound, not a formula for that minimum. Its purpose is an explicit synchronization proof. There is substantial room to shorten the construction.

The proof has three exact steps. First, p maps the whole state set onto a cycle and fixes that cycle pointwise. Second, the words ap and bp act on cycle indices as two almost matching reflections. Their discrepancy yields an elementary merger e . Third, a rotation followed by this merger repeatedly reduces the image until only one state remains. All three assertions are proved below for an arbitrary parameter.

2 Words, pair merging, and regular languages

2.1 Three different length parameters

Let $\mathcal{A} = (Q, \Sigma, \delta)$ be finite and complete, with $|Q| \geq 2$. For a nonempty subset $S \subseteq Q$ and a word w , write

$$Sw = \{qw : q \in S\}, \quad \text{rank}(w) = |Qw|.$$

The empty word ε acts as the identity. For $x, y \in Q$ define

$$\text{dist}(x, y) = \min\{|w| : xw = yw\}, \quad (2.1)$$

with value ∞ when the set is empty. The diagonal has distance zero. For a distinguished state q , the compress-with-another threshold is

$$\tau(q) = \min_{x \in Q \setminus \{q\}} \text{dist}(q, x). \quad (2.2)$$

The reset threshold is instead

$$\text{rt}(\mathcal{A}) = \min\{|w| : |Qw| = 1\}.$$

Finally, the pairwise merging diameter is $D(\mathcal{A}) = \max_{x, y \in Q} \text{dist}(x, y)$.

These parameters quantify different tasks. A word witnessing $\tau(q)$ may merge just one pair and leave almost every other state distinct. A reset word witnesses every pair's mergeability simultaneously. Whenever the reset threshold is finite,

$$\tau(q) \leq D(\mathcal{A}) \leq \text{rt}(\mathcal{A}). \quad (2.3)$$

The first inequality follows from a minimum being at most a maximum over a larger collection. The second follows because a reset word merges any chosen pair. Neither inequality usually identifies the other quantities exactly.

Lemma 2.1 (Pair criterion). *A finite complete automaton is synchronizing if and only if every pair of states can be merged by a word.*

Proof. A reset word merges every pair. Conversely, begin with the image set $S = Q$. If S has more than one element, choose distinct $x, y \in S$ and a word u with $xu = yu$. Applying u to every member of S cannot increase the image size and identifies at least two members, so $|Su| < |S|$. Repeat on the new image. Its positive integer size strictly decreases at each repetition, so after at most $|Q| - 1$ repetitions the concatenation of the chosen words has singleton image. $\square$

Our universal proof will not use finite pair enumeration. The lemma explains why the independently supplied pair certificates also verify synchronization on their finite parameter range.

2.2 The reset language and a right-linear grammar

The set

$$\text{Reset}(\mathcal{A}) = \{w \in \Sigma^* : |Qw| = 1\}$$

is regular. Indeed, consider a deterministic automaton whose states are the nonempty subsets of Q , whose initial state is Q , whose letter action is $S \mapsto Sx$, and whose accepting states are the singleton subsets. Induction on word length shows that its state after reading w is exactly Qw . This proves the claim without any minimization assumption.

It is also a two-sided ideal of the free monoid: if w is a reset word and u, v are arbitrary words, then

$$Q(uvw) = ((Qu)w)v \subseteq (Qw)v,$$

and the right-hand side is a singleton. Completeness ensures the left-hand side is nonempty, hence a singleton too.

For an explicitly grammatical description, introduce a nonterminal X_S for each nonempty reachable subset, with start symbol X_Q . For each letter x include the production $X_S \rightarrow xX_{Sx}$, and include $X_S \rightarrow \varepsilon$ exactly when S is a singleton. A derivation tracking its successive subscripts shows that this right-linear grammar generates precisely $\text{Reset}(\mathcal{A})$. Thus [theorem 1.1](#) gives a parameterized word in a well-defined regular language. The grammar observation is elementary background, not a new grammar-size theorem.

3 The exact automaton model

3.1 A total transition specification

Fix $k \geq 1$ and put

$$n = 3k + 6, \quad M = 2k + 4, \quad N = 3k + 5, \quad Q = \{q_0, q_1, \dots, q_N\}. \quad (3.1)$$

It is useful to call $q_0, \dots, q_M$ the core, q_M the junction, and $q_{M+1}, \dots, q_N$ the tail. The tail has $k + 1$ states.

Definition 3.1 (The figure-complete family). The automaton $\mathcal{A}_k$ has alphabet $\{a, b\}$ and the following actions:

$$q_i a = \begin{cases} q_0, & i = 0, \\ q_{i+1}, & 1 \leq i < M, \text{ } i \text{ odd}, \\ q_{i-1}, & 1 \leq i < M, \text{ } i \text{ even}, \\ q_{M+1}, & i = M, \\ q_3, & i = M + 1, \\ q_{\min(i+1, N)}, & M + 2 \leq i \leq N, \text{ } i \text{ even}, \\ q_{i-1}, & M + 2 \leq i \leq N, \text{ } i \text{ odd}, \end{cases} \quad (3.2)$$

$$q_i b = \begin{cases} q_{i+1}, & 0 \leq i < M, \text{ } i \text{ even}, \\ q_{i-1}, & 0 \leq i < M, \text{ } i \text{ odd}, \\ q_2, & i = M, \\ q_{\min(i+1, N)}, & M + 1 \leq i \leq N, \text{ } i \text{ odd}, \\ q_{i-1}, & M + 1 \leq i \leq N, \text{ } i \text{ even}. \end{cases} \quad (3.3)$$

Every pair (q_i, x) is covered by exactly one clause of the appropriate formula. The minimum in each outward tail rule is simply a compact way of writing the endpoint self-loop. In particular,

$$\begin{array}{c|cc} & q_N a & q_N b \\ \hline N \text{ even} & q_N & q_{N-1} \\ N \text{ odd} & q_{N-1} & q_N \end{array} \quad \text{and} \quad q_0 b = q_1. \quad (3.4)$$

The survey's printed cases omit $q_0 b$ and one endpoint transition; its Figure 3 supplies them. Equations (3.2)–(3.4) use that figure-complete interpretation, also used in [2]. This convention is essential: the theorem is not about an arbitrary completion of an incomplete transition table. For example, changing $q_0 b$ to q_0 would make the single letter b merge q_0 with q_1 .

3.2 Two immediate structural checks

Proposition 3.2. *The automaton $\mathcal{A}_k$ is strongly connected. Both letter transformations have rank $n - 1$. The only distinct-state collisions of one letter are*

$$\{q_4, q_{M+1}\}a = \{q_3\}, \quad \{q_3, q_M\}b = \{q_2\}.$$

Proof. On $q_0, \dots, q_{M-1}$, consecutive states are joined in both directions: b exchanges the pairs with even left endpoint, and a exchanges the pairs with odd left endpoint. Thus that segment is mutually reachable. From q_{M-1} an a reaches q_M , and from q_M a b returns to q_2 . The transition $q_M a = q_{M+1}$ reaches the tail; consecutive tail states are again joined in both directions. Finally $q_{M+1} a = q_3$ returns to the core. These routes prove strong connectivity.

The b -action consists of disjoint transpositions on the core segment, disjoint tail transpositions and one possible endpoint fixed point, together with $q_M b = q_2$. The latter duplicates the image of q_3 ; every other image has just one preimage, except the missing image q_M . Hence b has rank $n - 1$ with the stated collision. For a , the exceptional chain is

$$q_{M-1} \mapsto q_M \mapsto q_{M+1} \mapsto q_3.$$

All remaining arrows are disjoint transpositions or fixed points. Its final image q_3 is also the image of q_4 , while q_{M-1} is absent from the image. This gives the other assertion. $\square$

Strong connectivity does not itself prove synchronizability. Nor does the fact that both letters have rank $n - 1$. The comparison family in section 7 will show how a nontrivial permutation quotient can survive a very similar collection of arrows.

4 The cyclic retract

The main simplification appears after grouping letters into blocks ab . Use the reparameterization

$$r = k + 4 \geq 5, \quad M = 2r - 4, \quad N = 3r - 7, \quad n = 3r - 6. \quad (4.1)$$

Define r distinct states

$$s_0 = q_0, \quad s_j = q_{2j-1} \ (1 \leq j \leq r-2), \quad s_{r-1} = q_2, \quad C = \{s_0, \dots, s_{r-1}\}. \quad (4.2)$$

Whenever a subscript on s is interpreted modulo r , we say so or use an expression already in $\mathbb{Z}/r\mathbb{Z}$.

Lemma 4.1 (One recurrent cycle). *For $c = ab$, the action on C is*

$$s_j c = s_{j+1 \bmod r}. \quad (4.3)$$

Every state of Q enters C after at most $r - 2$ applications of c . The junction q_M enters C for the first time at q_2 after exactly $r - 2$ applications.

Proof. The beginning and end of the proposed cycle follow directly from the transition rules:

$$q_0 ab = q_1, \quad q_2 ab = q_0, \quad q_{M-1} ab = q_2.$$

For odd i with $1 \leq i \leq M - 3$, the two letters send q_i first to q_{i+1} and then to q_{i+2} . This proves the complete cycle

$$q_0 \mapsto q_1 \mapsto q_3 \mapsto \cdots \mapsto q_{M-1} \mapsto q_2 \mapsto q_0.$$

The remaining even core states obey

$$q_{2j} c = q_{2j-2} \quad (2 \leq j \leq r - 3). \quad (4.4)$$

Thus q_{2j} reaches q_2 in $j - 1 \leq r - 4$ blocks.

It remains to analyze the junction and tail. Let E be the largest even integer at most N , and O the largest odd integer at most N . The c -trajectory from the junction is

$$\begin{aligned} q_M &\mapsto q_{M+2} \mapsto \cdots \mapsto q_E \\ &\mapsto q_O \mapsto \cdots \mapsto q_{M+1} \mapsto q_2. \end{aligned} \quad (4.5)$$

The even portion lists the even tail states in increasing order, and the odd portion lists the odd tail states in decreasing order; an endpoint is listed only once when its portion has one state. Away from the endpoint, the even tail indices increase by two under c and the odd tail indices decrease by two. At the endpoint there are two cases. If N is even, $q_N a = q_N$ and $q_N b = q_{N-1}$, so $q_E c = q_O$. If N is odd, $E = N - 1$ and $q_E a = q_N$, $q_N b = q_N$, again giving $q_E c = q_O$. At the exit, $q_{M+1} a = q_3$ and $q_3 b = q_2$. This proves every arrow in (4.5).

The displayed trajectory visits every one of the $N - M = r - 3$ tail states exactly once: first its even indices, then its odd indices. None belongs to C . It therefore has $r - 2$ edges from q_M to its first point in C . Every tail state is already on this trajectory and needs fewer edges. Together with (4.4), this accounts for all of Q . $\square$

The recurrent behavior has become completely regular despite the parity-dependent endpoint. The parity changes only which of the last two states is visited first in the folded tail trajectory.

Lemma 4.2 (Idempotent retraction). *The word $p = c^r$ satisfies*

$$Qp = C, \quad s_j p = s_j \quad (0 \leq j < r), \quad p^2 = p \quad \text{as transformations on } Q. \quad (4.6)$$

Moreover,

$$q_{2j} p = s_{r-j} \quad (1 \leq j \leq r - 3), \quad (4.7)$$

$$q_M p = s_1, \quad (4.8)$$

$$q_{M+1} p = s_{r-2}. \quad (4.9)$$

Proof. Every state enters C by time $r - 2$, hence certainly by time r . On C , the word c^r is one complete turn around a cycle of length r , so it is the identity. These facts give both $Qp = C$ and $p^2 = p$.

For (4.7), q_{2j} reaches $q_2 = s_{r-1}$ after $j - 1$ blocks, including the case $j = 1$ in which it is there initially. The remaining $r - j + 1$ blocks put it at index

$$(r - 1) + (r - j + 1) \equiv r - j \pmod{r}.$$

For the junction, lemma 4.1 gives $q_M c^{r-2} = q_2$. The two remaining blocks send q_2 to q_0 and then $q_1 = s_1$. Finally $q_{M+1}c = q_2$; applying the remaining $r - 1$ blocks gives index $(r - 1) + (r - 1) \equiv r - 2 \pmod{r}$. $\square$

It is important that p fixes C pointwise; a word merely mapping Q into C would not justify the same subsequent formulas. It is equally important that c^{r-1} is an inverse of c only on C . The transformation c is not a permutation of the entire state set. No global inverse is used anywhere in the construction.

5 Two projected reflections and one elementary merger

A word v mapping C into itself induces a map on indices,

$$\bar{v} : \mathbb{Z}/r\mathbb{Z} \longrightarrow \mathbb{Z}/r\mathbb{Z}, \quad s_j v = s_{\bar{v}(j)}.$$

By lemma 4.2, both $F = ap$ and $H = bp$ have this property, even though a and b individually can leave C .

Lemma 5.1 (Projected letters). *The induced maps of $F = ap$ and $H = bp$ are*

$$\bar{F}(j) = \begin{cases} 1, & j = r - 2, \\ -j \pmod{r}, & j \neq r - 2, \end{cases} \quad \bar{H}(j) = 1 - j \pmod{r}. \quad (5.1)$$

Proof. For F , $s_0 a = q_0$ and $q_0 p = s_0$, giving $\bar{F}(0) = 0$. For $1 \leq j \leq r - 3$, we have $s_j = q_{2j-1}$ and $s_j a = q_{2j}$; (4.7) then gives $s_j a p = s_{r-j}$. At $j = r - 2$, $s_j = q_{M-1}$, so $s_j a = q_M$ and (4.8) gives $s_j a p = s_1$. This is the single exceptional value. At $j = r - 1$, $s_j = q_2$, $q_2 a = q_1 = s_1$, and p fixes s_1 . Thus the formula for F covers every index.

For H , the first two cases are $s_0 b = q_1 = s_1$ and $s_1 b = q_0 = s_0$. For $2 \leq j \leq r - 2$,

$$s_j b = q_{2j-2} = q_{2(j-1)}, \quad q_{2(j-1)} p = s_{r-j+1},$$

by (4.7). Finally $s_{r-1} = q_2$ and $q_2 b = q_3 = s_2$. These cases give $\bar{H}(j) = 1 - j$ modulo r throughout. $\square$

The map $\bar{H}$ is a genuine reflection of the cyclic indices. The map $\bar{F}$ would be the reflection $j \mapsto -j$ but for one value: index $r - 2$ is sent to 1 instead of 2. This one-unit defect is precisely what permits a reduction in rank.

Lemma 5.2 (An elementary merger). *The word*

$$e = FHc^{r-1} = apbp c^{r-1}$$

fixes every state of C except s_{r-2} , and sends s_{r-2} to s_{r-1} . Equivalently,

$$\bar{e}(j) = \begin{cases} r - 1, & j = r - 2, \\ j, & j \neq r - 2. \end{cases} \quad (5.2)$$

Proof. For a nonexceptional index j , applying $\bar{F}$ and then $\bar{H}$ gives $j \mapsto -j \mapsto 1 + j$. The final c^{r-1} subtracts one on C , so the result is j . For $j = r - 2$, the successive indices are

$$r - 2 \xrightarrow{F} 1 \xrightarrow{H} 0 \xrightarrow{c^{r-1}} r - 1.$$

This is exactly (5.2). $\square$

Although F and H are long words in the original alphabet, their restrictions are very small algebraic objects. The use of a retraction makes their compositions legitimate: after each projected action the current state is back in C . There is no assumption that the same formulas hold outside C .

6 From a single merger to a reset word

6.1 A general cycle-sweeping lemma

The final step is useful independently of the particular automaton.

Lemma 6.1 (Cycle and adjacent merger). *Let $C = \{s_0, \dots, s_{r-1}\}$ with $r \geq 3$. Suppose words c, e map C into itself, c advances indices by one modulo r , and e fixes all indices except an index t , which it sends to $t + 1$ modulo r . Put $h = ec^{r-1}$. Then*

$$\bar{h}(j) = \begin{cases} t, & j = t, \\ j - 1 \pmod{r}, & j \neq t, \end{cases} \quad (6.1)$$

and

$$|Ch^m| = \max(1, r - m) \quad (m \geq 0), \quad Ch^{r-2} = \{s_t, s_{t+1}\}, \quad Ch^{r-2}e = \{s_{t+1}\}. \quad (6.2)$$

If a word p maps an entire state set Q onto C , then $ph^{r-2}e$ resets Q .

Proof. At index t , e first advances to $t + 1$, and c^{r-1} returns to t . Every other index is fixed by e and then decremented by one. This proves (6.1).

List the cyclic indices in the order

$$t - 1, t - 2, \dots, t - (r - 1) = t + 1, t \quad \text{in } \mathbb{Z}/r\mathbb{Z}.$$

There are r distinct entries. The transformation h sends each entry to the next and fixes the final entry. After m iterations its image consists of the final $r - m$ entries when $m < r$, and just the last entry when $m \geq r - 1$. This proves the rank formula and the description of Ch^{r-2} . The map e sends both of its remaining indices to $t + 1$. Prepending p replaces the initial image Q by C , proving the last assertion. $\square$

6.2 Proof and length of the explicit construction

Proof of theorem 1.1. By lemma 4.2, the first factor p maps Q onto C . By lemma 5.2, e is the adjacent merger with exceptional index $t = r - 2$. Apply lemma 6.1: the suffix $(ec^{r-1})^{r-2}e$ maps C to $s_{r-1} = q_2$. Consequently $QR_k = \{q_2\}$.

For the length calculation, all exponents in the word expression are nonnegative integers and all products are literal concatenations. We have

$$|c| = 2, \quad |p| = 2r,$$

$$|e| = 1 + 2r + 1 + 2r + 2(r - 1) = 6r, \quad |ec^{r-1}| = 8r - 2.$$

Hence

$$\begin{aligned} |R_k| &= 2r + (r - 2)(8r - 2) + 6r \\ &= 8r^2 - 10r + 4. \end{aligned}$$

Substituting $r = k + 4$ gives $8k^2 + 54k + 92$. Substituting $r = n/3 + 2$ gives $8n^2/9 + 22n/3 + 16$. This proves every assertion of the theorem. $\square$

The state-set reduction is especially transparent:

$$Q \xrightarrow{p} C \xrightarrow{h^{r-2}} \{s_{r-2}, s_{r-1}\} \xrightarrow{e} \{s_{r-1}\}.$$

The word has a short description even when its expanded length is large. The implementation evaluates its transformations by repeated squaring, using $O(n \log r)$ elementary state-array operations and $O(n)$ working memory for the fixed number of arrays. Writing the expanded word instead requires $\Theta(r^2)$ output symbols. These are different computational tasks; the large-parameter checks use the compressed description.

6.3 The smallest case worked out completely

For $k = 1$, the state set is $\{q_0, \dots, q_8\}$ and $r = 5$. The two transition arrays, written as target indices, are

$$\begin{array}{c|cccccccc} i & 0 & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \\ \hline a(i) & 0 & 2 & 1 & 4 & 3 & 6 & 7 & 3 & 8 \\ b(i) & 1 & 0 & 3 & 2 & 5 & 4 & 2 & 8 & 7 \end{array} \quad (6.3)$$

The $c = ab$ cycle and its longest incoming trajectory are

$$0 \rightarrow 1 \rightarrow 3 \rightarrow 5 \rightarrow 2 \rightarrow 0, \quad 6 \rightarrow 8 \rightarrow 7 \rightarrow 2.$$

Thus the original state indices in the cycle order of C are $(0, 1, 3, 5, 2)$. On these five cycle positions, not on the original state numbering, the relevant transformations are

$$\begin{array}{c|ccccc} j & 0 & 1 & 2 & 3 & 4 \\ \hline \overline{F}(j) & 0 & 4 & 3 & 1 & 1 \\ \overline{H}(j) & 1 & 0 & 4 & 3 & 2 \\ \overline{e}(j) & 0 & 1 & 2 & 4 & 4 \\ \overline{h}(j) & 4 & 0 & 1 & 3 & 3 \end{array}$$

The chain under h is $2 \rightarrow 1 \rightarrow 0 \rightarrow 4 \rightarrow 3 \rightarrow 3$. Therefore $Ch^3 = \{s_4, s_3\}$ and $Ch^3e = \{s_4\} = \{q_2\}$. The word ph^3e has length $10 + 3 \cdot 38 + 30 = 154$. This example also illustrates why mixing state labels with cycle indices would lead to incorrect formulas.

7 Why one fewer tail state changes the answer

A nearby nonsynchronizing example helps identify what the reset proof actually exploits. It also gives a negative test for both the mathematics and the implementation.

Definition 7.1 (The shortened-tail family). Keep $k \geq 1$, $r = k + 4$, and $M = 2r - 4$, but use the state set

$$Q' = \{q_0, \dots, q_{N'}\}, \quad N' = 3r - 8 = 3k + 4.$$

Define $\mathcal{B}_k$ by the same formulas as (3.2) and (3.3), with N' in place of N . In particular the new endpoint has the same parity-based reflecting self-loop convention. The tail now has $k = r - 4$ states, and $|Q'| = 3k + 5$.

This operation is not simply restricting both old transformations to a subset: the new endpoint's outward arrow is replaced by a self-loop. For $k = 1$ the tail consists of the single state q_{M+1} ; its a -image is still the exceptional exit q_3 , and its b -image is itself.

Theorem 7.2 (A dihedral permutation quotient). *Every $\mathcal{B}_k$ is strongly connected but not synchronizing. More precisely, there is a surjection $\pi : Q' \rightarrow \mathbb{Z}/r\mathbb{Z}$ such that*

$$\pi(qa) = -\pi(q), \quad \pi(qb) = 1 - \pi(q) \quad (q \in Q'). \quad (7.1)$$

Proof. Strong connectivity follows by the same core and tail routes as in [proposition 3.2](#); the one-state tail causes no exception to those routes. Define the colors modulo r by

$$\begin{aligned} \pi(q_0) &= 0, \\ \pi(q_{2j-1}) &= j \quad (1 \leq j \leq r-2), \\ \pi(q_{2j}) &= r-j \quad (1 \leq j \leq r-2), \\ \pi(q_{M+2s+1}) &= r-2-s \quad (s \geq 0, M+2s+1 \leq N'), \\ \pi(q_{M+2s}) &= 2+s \quad (s \geq 1, M+2s \leq N'). \end{aligned} \quad (7.2)$$

These cases partition Q' . The states of C , which are all still present, have colors $0, 1, \dots, r-1$, so π is surjective.

On the ordinary core transpositions of a , the paired colors are j and $r-j$, which are negatives. The fixed state q_0 has color zero. On the exceptional a -chain the colors are

$$r-2 \mapsto 2 \mapsto r-2 \mapsto 2,$$

again changing sign at each step. A core b -pair with left endpoint q_{2j} , $j \geq 1$, has colors $r-j$ and $j+1$; these sum to 1 modulo r . The pair q_0, q_1 has colors 0, 1, and the junction transition $q_M b = q_2$ has colors 2, $r-1$, again summing to 1.

In the tail, an ordinary a -pair q_{M+2s}, q_{M+2s+1} has colors $2+s, r-2-s$, which are negatives. An ordinary b -pair q_{M+2s+1}, q_{M+2s+2} has colors $r-2-s, 3+s$, which sum to 1 modulo r . The first tail state's exceptional a -exit has colors $r-2, 2$, as required. Thus only the endpoint self-loop remains to check.

If r is even, N' is even and $N' - M = r-4$ is even. The endpoint color is $r/2$, fixed by $x \mapsto -x$, which is the letter a that self-loops. If r is odd, N' is odd and its color is $(r+1)/2$, fixed by $x \mapsto 1-x$, which is the self-looping letter b . The endpoint's other letter is already covered by a tail pair, or by the exceptional exit when $k=1$. This completes the verification of (7.1).

Both maps $x \mapsto -x$ and $x \mapsto 1-x$ are permutations of $\mathbb{Z}/r\mathbb{Z}$. By induction on word length, every word w induces a permutation g_w with $\pi(qw) = g_w(\pi(q))$. Consequently

$$\pi(Q'w) = g_w(\pi(Q')) = g_w(\mathbb{Z}/r\mathbb{Z}) = \mathbb{Z}/r\mathbb{Z}.$$

In particular $|Q'w| \geq r > 1$ for every word. No reset word exists. $\square$

Corollary 7.3 (A single endpoint defect). *Extend the same color formulas (7.2) to $\mathcal{A}_k$ by allowing its one additional endpoint. The two relations (7.1) hold at every state-letter pair except the self-loop at q_N .*

Proof. All non-self-loop verifications in the preceding proof remain valid. If r is even, then N is odd, its color is $r/2$, and its b -loop would need to send that color to $r/2+1$ instead of fixing it. If r is odd, then N is even, its color is $(r+1)/2$, and its a -loop would need to send that color to $(r-1)/2$. These colors differ modulo r since $r \geq 5$. $\square$

This explains the usefulness of the projected reflections. In the shorter family, a perfect reflection action survives as a quotient and forbids synchronization. In the actual family, precisely one local arrow violates that obstruction. The cyclic retraction turns the violation into an explicit adjacent merger. A vague assertion that “the letters almost act as permutations” would miss the decisive feature in either direction.

8 The short merging word: a known result, reproved

Let $\tau_k = \tau_{\mathcal{A}_k}(q_0)$. The word in the numerical part of the original conjecture has one letter more than its stated length. The corrected all-parameter upper bound is established in [2]; we include a short proof using the trajectory already proved above.

Proposition 8.1 (Known upper bound). *For every $k \geq 1$, the word*

$$w_k = (ba)^{k+2}(ab)^{k+2} \quad (8.1)$$

sends q_0 and q_2 to q_2 . Thus

$$\tau_k \leq 4k + 8 = \frac{4}{3}n. \quad (8.2)$$

Proof. Set $u = ba$, $c = ab$ and retain $r = k + 4$. On the even core states,

$$q_{2j}u = q_{2j+2} \quad (0 \leq j \leq r-3), \quad q_M u = q_1.$$

Hence $q_0 u^{r-2} = q_M$ and $q_2 u^{r-2} = q_1$. By [lemma 4.1](#), $q_M c^{r-2} = q_2$. The point $q_1 = s_1$ is on the c -cycle, so $q_1 c^{r-2} = s_{r-1} = q_2$. Concatenating the blocks proves both merging identities. Their combined length is $4(r-2) = 4k + 8$. $\square$

The elementary free-monoid identity $(ab)^m a = a(ba)^m$ gives

$$(ab)^{k+2} a (ab)^{k+2} = a w_k, \quad |a w_k| = 4k + 9. \quad (8.3)$$

Since $q_0 a = q_0$ and $q_1 a = q_2$, this longer word merges q_0 with q_1 . The length discrepancy is not a counterexample to the numerical bound $4k + 8$; it is a distinction between a displayed witness, its partner, and the minimum over all possible partners.

Conjecture 8.2 (Remaining numerical clause). *For every $k \geq 1$, $\tau_k = 4k + 8$.*

Only the lower inequality is missing after [proposition 8.1](#). The universal reset word in [theorem 1.1](#) does not establish it: a word that eventually merges *all* states supplies no reason why a *particular pair* could not merge earlier than the proposed minimum. The two questions must remain separate even though they originated in one conjecture.

9 Exact computations with independently checkable certificates

9.1 A finite graph of unordered pairs

For an n -state automaton, take all unordered pairs with repetition,

$$\mathcal{P} = \{\{q_i, q_j\} : 0 \leq i \leq j < n\}.$$

There are $n(n+1)/2$ such vertices, including n diagonal pairs. Each letter induces one outgoing edge from each vertex. The merging distance of a pair is exactly its directed distance to the diagonal set. This follows because a path labeled by w ends at $\{q_i w, q_j w\}$.

Reverse breadth-first search from all diagonal vertices computes these distances in $O(n^2)$ time and space for a fixed two-letter alphabet: the graph has $n(n+1)/2$ vertices and twice as many directed edges. The supplied generator stores the reverse adjacency lists in integer arrays. It writes one nonnegative integer for each pair. An unreachable pair would have value -1 in the generator, but the verifier would reject such a certificate as a certificate of complete mergeability.

9.2 Why the certificate proves an exact distance

Lemma 9.1 (Bellman certificate). *Let $\mathbb{N}$ include zero. Suppose a function $d : \mathcal{P} \rightarrow \mathbb{N}$ has value zero exactly on the diagonal and satisfies, for every nondiagonal pair P ,*

$$d(P) = 1 + \min\{d(Pa), d(Pb)\}. \quad (9.1)$$

Then $d(P)$ equals the shortest merging length of P for every P .

Proof. For either letter x , (9.1) implies $d(P) \leq 1 + d(Px)$. The same inequality holds on a diagonal because its value and the values of both successors are zero. Along any word of length ℓ that merges P , telescoping gives $d(P) \leq \ell + 0$. Thus $d(P)$ is a lower bound for every merging length.

If P is nondiagonal, one of the two successor values is exactly $d(P) - 1$. Choose such a letter and repeat while the value is positive. A nonnegative integer decreases by one at each step, so after exactly $d(P)$ steps it reaches zero. Zero occurs only on the diagonal. The resulting word merges P and has length $d(P)$, proving the matching upper bound. $\square$

This proof does not depend on how the values were discovered. In particular the verifier need not rerun breadth-first search or trust the generator's queue discipline. It checks only the finite table, the explicit transition model, and the displayed local identity.

9.3 Results actually verified

For every integer $1 \leq k \leq 200$, the supplied certificates verify

$$\tau_k = 4k + 8, \quad \text{dist}(q_0, q_1) = 4k + 9, \quad (9.2)$$

and identify the complete set of optimal partners of q_0 as $\{q_2, q_{2k+5}\}$. They also verify, on the same finite range,

$$D(\mathcal{A}_k) = \begin{cases} (5k^2 + 32k + 47)/4, & k \text{ odd,} \\ (5k^2 + 34k + 44)/4, & k \text{ even.} \end{cases} \quad (9.3)$$

The prior note [2] reports the threshold checks through $k = 60$ and the diameter formula through $k = 16$. Here all values are independently recomputed and checked, not copied from that source.

k	n	τ_k	$\text{dist}(q_0, q_1)$	$D(\mathcal{A}_k)$	$ R_k $
1	9	12	13	21	154
2	12	16	17	33	232
3	15	20	21	47	326
4	18	24	25	65	436
5	21	28	29	83	562
10	36	48	49	221	1432
20	66	88	89	681	4372
40	126	168	169	2351	15052
60	186	248	249	5021	32132
100	306	408	409	13361	85492
150	456	608	609	29411	188192
200	606	808	809	51711	330892

Table 1: Selected certified pair-distance results. The last column is the length of the universal construction, not a computed shortest reset length. The archive contains every intervening parameter, not only these rows.

The largest automaton in the exhaustive pair calculation has 606 states and 183,921 pair vertices. Across the 200 certificates there are 12,486,300 pair entries, each checked against either the diagonal condition or the Bellman identity. All entries are finite and all the stated identities passed.

Separately, every structural identity in the universal construction was checked for $k = 1, \dots, 300$ and at $k = 500, 1000, 10000$. Those tests include the retraction image, the two projected maps, the elementary merger, selected ranks of powers of the sweeping map, and the final singleton image. They also check the shortened-family quotient and the unique endpoint defect. For $k = 1, \dots, 10$, the reset words were expanded and evaluated both as compositions of state arrays and by direct state-by-state simulation.

These computations support implementation fidelity and help catch indexing errors. The proofs in [sections 4 to 7](#) are the justification for universal statements. Testing the word at $k = 10000$ is not being presented as a substitute for induction or a symbolic argument.

9.4 Implementation independence and negative controls

The generator and verifier build the transition functions differently. The generator uses the piecewise rules. The verifier initializes identity maps, inserts disjoint transpositions, and then inserts the exceptional junction arrows. It does not import the generator or its constructor. The structural test program compares the two constructions directly.

The verifier checks every entry using ordinary Python integers and explicit exceptions. Its safety conditions are not written as `assert` statements that disappear under `python -O`. It also rejects three deliberately corrupted versions of the smallest certificate: a nonzero diagonal, an incremented nondiagonal distance, and a nondiagonal distance replaced by zero. Random-word tests with a fixed seed check the left-to-right composition convention against a second evaluator.

This is an independent executable checker in the algorithmic sense, not a separately developed or proof-assistant-verified implementation. A shared misreading of the source figure would still be a possible error unless the source-model correspondence were checked; this is why the complete transition specification and endpoint discussion are included in the article itself.

10 What the argument does not yet prove

10.1 The universal lower bound

To finish [conjecture 8.2](#), it would suffice to produce, for arbitrary k , a nonnegative potential L_k on unordered pairs such that it vanishes on the diagonal and satisfies

$$L_k(P) \leq 1 + L_k(Pa), \quad L_k(P) \leq 1 + L_k(Pb), \quad (10.1)$$

together with

$$L_k(\{q_0, q_j\}) \geq 4k + 8 \quad (1 \leq j \leq N). \quad (10.2)$$

The telescoping half of [lemma 9.1](#) would then imply the missing lower bound. Equality in a Bellman equation would not be necessary for this purpose. A piecewise formula proving only the lower bound might be simpler than a formula for all exact distances.

The finite distance tables provide examples of such potentials for the checked parameters, but this article does not extract a universally valid formula from them. Nor does it give a finite parameter-independent partition of all pair configurations with verified affine transition inequalities. Those are precise missing steps, not consequences of the reset construction.

10.2 Why the synchronization proof cannot be reversed

The retraction p deliberately forgets transient position. Many states outside C acquire the same projected state, and that is useful for constructing a reset word. For a shortest-word lower bound, however, forgetting transient position can erase the very time cost one needs to measure. Showing that a particular long word collapses a projected cycle does not imply that every shorter word fails to merge the distinguished state with some partner.

Likewise, the dihedral coloring in [section 7](#) is an exact obstruction only for the shortened family. In $\mathcal{A}_k$ its one defective self-loop changes the color relation. A lower-bound argument based on that coloring would have to track both when a trajectory reaches the defect and what transient information remains after leaving it. Treating all letters as reflections throughout a proposed short word would be false precisely at that critical transition.

A further source of misleading lower bounds is symmetrizing the pair graph. Allowing a transition to be traversed backward can only shorten distances, so undirected distance is a legitimate relaxation but need not be sharp. One must not replace a directed shortest-path problem by a symmetric geometric picture and then silently retain the claimed exact length.

10.3 Other conjectural patterns and distinct neighboring questions

The formula (9.3) is certified here only through $k = 200$. Its algebraic leading term would be $(5/36)n^2$ after substituting $n = 3k + 6$, but finite agreement does not establish an asymptotic theorem for $D(\mathcal{A}_k)$. The same limitation applies to the observed optimal-partner classification outside the certified range.

Triple rendezvous time asks for a shortest word merging *some three* initial states. A recent preprint by Zhang [3] gives exact $\lfloor 4n/3 \rfloor$ values for a different family. This is a neighboring invariant, not the distinguished-state threshold of $\mathcal{A}_k$, and not the universal reset problem solved by our construction. Coinciding numerical expressions do not identify the automata or the quantified tasks.

Finally, the quadratic bound (1.3) is not presented as optimal. The construction spends many letters restoring states to the cyclic retract after each projected action. Shorter projected words or more economical sweeps could lower it. The exact reset threshold itself has not been computed by the pair-distance procedure: it would require an appropriate subset calculation or another proof.

11 Conclusion

The selected open synchronization question has an explicit answer for the figure-complete family: every $\mathcal{A}_k$ resets under

$$R_k = p(ec^{r-1})^{r-2}e, \quad c = ab, \quad p = c^r, \quad e = apbpc^{r-1}, \quad r = k + 4.$$

The mechanism is a cyclic retraction, two projected reflections differing at one index, and an elementary adjacent merger. It is uniform in the parameter and includes both endpoint parities without an appeal to experimental extrapolation.

The shortened-tail family supplies a matching structural warning: restoring a perfect dihedral quotient prevents synchronization altogether. This comparison identifies the endpoint defect rather than strong connectivity or near-involutivity as the feature used by the proof.

The larger numerical conjecture remains a separate problem. Its upper bound is prior work and is reproved here; its lower bound is independently certified through 606 states, but not proved for arbitrary k . The archive is intended to make both the affirmative theorem and that boundary of knowledge easy to inspect and reproduce.

A Reproducing the artifacts

All Python programs use only the standard library and require Python 3.10 or newer. From the archive’s top directory, run

```
python code/test_structure.py
python code/verify_certificates.py --max-k 200
```

The second command checks the existing certificates without regenerating them. To reproduce the data and CSV summary, run

```
python code/generate_certificates.py --max-k 200
python code/verify_certificates.py --max-k 200
```

On a constrained execution service the generator may instead be split into two runs:

```
python code/generate_certificates.py --max-k 150
python code/generate_certificates.py --min-k 151 --max-k 200
```

The first run replaces the summary and the second appends to it. Do not append the same range twice without regenerating the summary from $k = 1$.

The certificate for parameter k is `certificates/k_XXXX.json.gz`, where `XXXX` is k padded to four decimal digits. Its distance array uses index

$$I(i, j) = \frac{\max(i, j)(\max(i, j) + 1)}{2} + \min(i, j).$$

Thus pairs are listed by increasing larger endpoint and then by increasing smaller endpoint. The verifier derives transitions independently from k ; transition arrays supplied by the generator are not trusted or needed.

The file `results/summary.csv` contains one row for each parameter. The three text logs record data generation, certificate checking, and structural testing. These logs report completed runs, not predicted results. The directory `code/` includes a guarded expansion routine for small reset words and a transformation-based evaluation routine for large ones. The file `PROOF_AUDIT.md` maps every main claim to its proof, computational status, and relevant limitation.

To rebuild the document with a standard TeX Live installation:

```
pdflatex -halt-on-error article.tex
pdflatex -halt-on-error article.tex
pdflatex -halt-on-error article.tex
```

The bibliography is embedded below, so no bibliography database or network access is needed to compile. A separate `references.bib` is supplied for reuse, not required by this build.

B A compact algebraic audit

For checking the proof without following the discussion in its original order, the exact dependencies are as follows. The transition formulas imply the cycle and tail trajectory in [lemma 4.1](#). Their lengths imply $p = c^r$ is an idempotent retraction, together with the three projection identities of [lemma 4.2](#). Those identities alone determine the actions of ap and bp in [lemma 5.1](#). Composing those two maps and the inverse cycle rotation on C yields the one-point merger of [lemma 5.2](#). The path argument in [lemma 6.1](#) then proves the reset identity. No computational assertion is an input to this chain.

There are three conventions on which the argument is especially sensitive. First, the letter a is applied *before* b in $c = ab$. Second, $q_0b = q_1$ and the nonloop endpoint arrow returns to the previous state. Third, indices modulo r label the ordered states s_j of the retract; they are not the original labels on q_i . Each convention is exposed in both the formulas and the executable examples.

References

- [1] Marek Szykuła. Synchronizing Automata: Open Problems. *Electronic Proceedings in Theoretical Computer Science*, 451:33–47, 2026. DOI: [10.4204/EPTCS.451.3](https://arxiv.org/abs/2608.24245). <https://arxiv.org/abs/2608.24245>. Version 1, 25 August 2026; in particular Section 2.3, Figure 3 and Conjecture 8. Consulted 20 September 2026.
- [2] Korea Superintelligence Labs; generated by Machina Mathematica. The compress-with-another threshold of the series A_k : an upper bound $4n/3$ for every k , exact values to $n = 186$, and a one-letter correction. *Ideosphere*, version 1, source snapshot 7 September 2026. <https://ideosphere.ai/papers/cwa-series>. Archived manuscript: <https://ideosphere.ai/papers/cwa-series-v1.pdf>. AI-generated research manuscript; independent human review listed as open. Consulted 20 September 2026. Its reported Lean build was not rerun for the present article.
- [3] Enkai Zhang. The triple rendezvous time of a synchronizing automaton can be $\lfloor 4n/3 \rfloor$. Preprint, arXiv:2609.19173v1, 14 September 2026. <https://arxiv.org/abs/2609.19173>. Consulted 20 September 2026. This paper concerns a different family and is cited only to distinguish the neighboring invariant.